Mathematical Analysis of the Cone Surface

Complete Differential-Geometric Treatment:
Curvature, Surface Area, Volume, and Geometric Properties

Dogan Balban

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Abstract

This paper presents a complete mathematical analysis of the cone surface defined by

$$z = -\frac{1}{12} \cdot 3^2 + 6 \cdot \sqrt{\frac{x^2}{\left(\frac{12}{\pi}\right)^2} + \frac{y^2}{\left(\frac{12}{\pi}\right)^2}}.$$

We investigate the algebraic structure, geometric properties, differential geometry, curvature behaviour, and surface parametrization. The analysis covers partial derivatives, gradient, normal vectors, surface area, volume, and a complete characterisation of this cone.

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1 Introduction and Simplification

1.1 Given Expression

We consider the surface defined by:

$$z = -\frac{1}{12} \cdot 3^2 + 6 \cdot \sqrt{\frac{x^2}{\left(\frac{12}{\pi}\right)^2} + \frac{y^2}{\left(\frac{12}{\pi}\right)^2}} \tag{1}$$

1.2 Algebraic Simplification

Proposition 1.1 (Simplification)**.** *Expression* (1) *simplifies to:*

$$z = -\frac{3}{4} + \frac{\pi}{2}\sqrt{x^2 + y^2}. \tag{2}$$

Proof. **Step 1** (Constant term): $-\frac{1}{12} \cdot 3^2 = -\frac{9}{12} = -\frac{3}{4}$.

Step 2 (Radicand): $\dfrac{x^2}{(12/\pi)^2} = x^2 \cdot \dfrac{\pi^2}{144}$, and analogously for y.

Step 3: $\sqrt{\dfrac{\pi^2}{144}(x^2 + y^2)} = \dfrac{\pi}{12}\sqrt{x^2 + y^2}$.

Step 4: $6 \cdot \dfrac{\pi}{12}\sqrt{x^2 + y^2} = \dfrac{\pi}{2}\sqrt{x^2 + y^2}$.

Step 5: Combining: $z = -\frac{3}{4} + \frac{\pi}{2}\sqrt{x^2 + y^2}$. □

1.3 Cylindrical Coordinates

Definition 1.1 (Cylindrical form)**.** Setting $x = r\cos\theta$, $y = r\sin\theta$, $r = \sqrt{x^2 + y^2}$, the equation reduces to:

$$\boxed{z = -\frac{3}{4} + \frac{\pi}{2}r.} \tag{3}$$

Remark 1.1. Equation (3) is independent of θ: the surface is a **surface of revolution** about the z-axis – specifically, a **right circular cone**.

2 Geometric Characterisation

2.1 Identification as a Cone

Theorem 2.1 (Cone property)**.** *The surface defined by* (3) *is a right circular cone with:*

- *apex at* $(0, 0, -\frac{3}{4})$,
- *axis along the positive* z*-axis,*
- *slope* $m = \frac{\pi}{2}$.

Proof. The equation $z = -\frac{3}{4} + \frac{\pi}{2}r$ is linear in r with positive coefficient. At $r = 0$ we have $z = -\frac{3}{4}$ (apex). For $r > 0$, z increases linearly – the characteristic property of a right circular cone. □

2.2 Cone Angles

Definition 2.1 (Cone half-angle)**.** The *cone half-angle* α is the angle between the cone axis (z-axis) and a generator (slant line).

Proposition 2.2 (Cone half-angle)**.** *For our cone:*

$$\alpha = \arctan\left(\frac{2}{\pi}\right) \approx 0.5669 \ rad \approx 32.48^\circ. \tag{4}$$

Proof. A generator runs from (r_0, θ_0, z_0) to $(r_0 + \Delta r, \theta_0, z_0 + \Delta z)$ with $\Delta z = \frac{\pi}{2}\Delta r$. The angle *between the axis and the generator* satisfies:

$$\tan\alpha = \frac{\Delta r}{\Delta z} = \frac{1}{\pi/2} = \frac{2}{\pi} \quad \Rightarrow \quad \alpha = \arctan\left(\frac{2}{\pi}\right) \approx 32.48^\circ.$$

□

Remark 2.1 (Elevation angle and complementary angle). The complementary *elevation angle* (angle between generator and the horizontal xy-plane) is:

$$\beta = \arctan\left(\frac{\pi}{2}\right) \approx 1.0039 \text{ rad} \approx 57.52^\circ,$$

with $\alpha + \beta = \frac{\pi}{2}$ ($32.48^\circ + 57.52^\circ = 90^\circ$). Some literature uses β as the "cone angle"; in this paper α always denotes the angle between axis and generator (Definition 2.1).

Corollary 2.3 (Apex angle)**.** *The full apex angle (between two opposite generators) is:*

$$2\alpha = 2\arctan\left(\frac{2}{\pi}\right) \approx 64.96^\circ.$$

2.3 Cross-Sections

Proposition 2.4 (Base circle)**.** *The cone intersects $z = 0$ in a circle of radius $r_0 = \frac{3}{2\pi} \approx 0.4775$.*

Proposition 2.5 (Horizontal cross-section)**.** *The intersection with $z = h$ (for $h \geq -\frac{3}{4}$) is a circle of radius:*

$$r(h) = \frac{2}{\pi}\left(h + \frac{3}{4}\right). \tag{5}$$

3 Functional Representation

3.1 Explicit Form and Partial Derivatives

As a graph: $f(x, y) = -\frac{3}{4} + \frac{\pi}{2}\sqrt{x^2 + y^2}$, $(x, y) \in \mathbb{R}^2$.

Proposition 3.1 (Partial derivatives)**.** *For $(x, y) \neq (0, 0)$:*

$$f_x = \frac{\pi x}{2r}, \quad f_y = \frac{\pi y}{2r}; \tag{6}$$

$$f_{xx} = \frac{\pi y^2}{2r^3}, \quad f_{yy} = \frac{\pi x^2}{2r^3}, \quad f_{xy} = -\frac{\pi xy}{2r^3}. \tag{7}$$

Remark 3.1. At the apex $(0, 0)$ the partial derivatives are undefined: this is the geometric singularity of the cone.

3.2 Gradient and Normal Vector

Proposition 3.2 (Gradient and normal).

$$\nabla f = \left(\frac{\pi x}{2r},\ \frac{\pi y}{2r}\right), \quad |\nabla f| = \frac{\pi}{2} \quad (\textit{constant for all } (x,y) \neq (0,0)),$$
$$\vec{n} = \left(\frac{\pi x}{2r},\ \frac{\pi y}{2r},\ -1\right), \quad \hat{n} = \frac{1}{\sqrt{1+\pi^2/4}}\vec{n}.$$

4 Parametrization

Definition 4.1 (Parametrization). A natural parametrization of the cone surface is:

$$\vec{\gamma}(r,\theta) = \left(r\cos\theta,\ r\sin\theta,\ -\tfrac{3}{4}+\tfrac{\pi}{2}r\right)^\top, \quad r \geq 0,\ \theta \in [0, 2\pi). \tag{8}$$

Proposition 4.1 (Tangent vectors and cross product).

$$\frac{\partial\vec{\gamma}}{\partial r} = (\cos\theta,\ \sin\theta,\ \tfrac{\pi}{2})^\top, \quad \frac{\partial\vec{\gamma}}{\partial\theta} = (-r\sin\theta,\ r\cos\theta,\ 0)^\top,$$
$$\frac{\partial\vec{\gamma}}{\partial r} \times \frac{\partial\vec{\gamma}}{\partial\theta} = \left(-\tfrac{\pi r}{2}\cos\theta,\ -\tfrac{\pi r}{2}\sin\theta,\ r\right)^\top,$$
$$\left|\frac{\partial\vec{\gamma}}{\partial r} \times \frac{\partial\vec{\gamma}}{\partial\theta}\right| = r\sqrt{1+\frac{\pi^2}{4}}.$$

Proof. Direct computation. Magnitude: $\sqrt{(\pi r/2)^2(\cos^2\theta+\sin^2\theta)+r^2} = r\sqrt{1+\pi^2/4}$. □

5 Surface Area

Theorem 5.1 (Lateral surface area of frustum). *The lateral surface area of the cone frustum between radii r_1 and r_2 ($0 \leq r_1 < r_2$) is:*

$$A = \pi\sqrt{1+\frac{\pi^2}{4}} \cdot (r_2^2 - r_1^2). \tag{9}$$

Proof. $A = \int_0^{2\pi}\int_{r_1}^{r_2} \sqrt{1+\pi^2/4}\cdot r\,dr\,d\theta = \sqrt{1+\pi^2/4}\cdot 2\pi\cdot\frac{r_2^2-r_1^2}{2} = \pi\sqrt{1+\pi^2/4}\cdot(r_2^2-r_1^2)$. □

Corollary 5.2 (Surface area from apex). *The lateral surface area from the apex to radius r:*

$$A(r) = \pi r^2\sqrt{1+\frac{\pi^2}{4}} \approx 1.8621 \cdot \pi r^2. \tag{10}$$

6 Volume

Theorem 6.1 (Volume of frustum). *The volume of the frustum between heights h_1 and h_2 ($-\frac{3}{4} \leq h_1 < h_2$) is:*

$$V = \frac{4}{3\pi}\left[\left(h_2+\frac{3}{4}\right)^3 - \left(h_1+\frac{3}{4}\right)^3\right]. \tag{11}$$

Proof. Using $r(h) = \frac{2}{\pi}(h + \frac{3}{4})$ from Theorem 2.5:

$$V = \int_{h_1}^{h_2} \pi \left[r(h)\right]^2 dh = \frac{4}{\pi} \int_{h_1}^{h_2} \left(h + \frac{3}{4}\right)^2 dh = \frac{4}{3\pi} \left[\left(h + \frac{3}{4}\right)^3\right]_{h_1}^{h_2}.$$

Verification: For $h_1 = -\frac{3}{4}$, $h_2 = h$, the classical cone formula $V = \frac{1}{3}\pi R^2 H$ with $R = r(h)$, $H = h + \frac{3}{4}$ gives the same result. □

7 Curvature Analysis

7.1 Gaussian Curvature

Theorem 7.1 (Gaussian curvature)**.** *At every regular point of the cone:*

$$K = 0.$$

Proof. A cone is a *developable surface* (locally isometric to the plane): $K = 0$. Directly: $\kappa_1 = 0$ along generators (straight lines), so $K = \kappa_1\kappa_2 = 0$. □

7.2 Mean Curvature

Proposition 7.2 (Mean curvature)**.** *At a point with radial coordinate $r > 0$:*

$$H = \frac{\pi}{4r\sqrt{1 + \frac{\pi^2}{4}}} = \frac{\pi}{2r\sqrt{4 + \pi^2}}. \tag{12}$$

Proof. For $z = f(x, y)$, the mean curvature is:

$$H = \frac{(1 + f_y^2)f_{xx} - 2f_x f_y f_{xy} + (1 + f_x^2)f_{yy}}{2(1 + f_x^2 + f_y^2)^{3/2}}.$$

Since $f_x^2 + f_y^2 = \pi^2/4$ (constant, Theorem 3.2), the denominator is $2(1 + \pi^2/4)^{3/2}$. Expanding the numerator using Theorem 3.1:

$$\text{Numerator} = \frac{\pi y^2}{2r^3} + \frac{\pi x^2}{2r^3} + \frac{\pi^3(x^4 + y^4 + 2x^2y^2)}{8r^5} = \frac{\pi}{2r} + \frac{\pi^3}{8r} = \frac{\pi(4 + \pi^2)}{8r}.$$

Hence:

$$H = \frac{\pi(4 + \pi^2)/8r}{2[(4 + \pi^2)/4]^{3/2}} = \frac{\pi}{2r\sqrt{4 + \pi^2}} = \frac{\pi}{4r\sqrt{1 + \pi^2/4}}. \quad \square$$

□

Remark 7.1. Equivalently, using the parametric formula $H = m/(2r\sqrt{1 + m^2})$ with slope $m = \pi/2$ yields $H = (\pi/2)/(2r\sqrt{1 + \pi^2/4}) = \pi/(4r\sqrt{1 + \pi^2/4})$ directly. Since $\kappa_1 = 0$, the second principal curvature is $\kappa_2 = 2H$.

8 Geodesics

Theorem 8.1 (Geodesics on the cone)**.** *The geodesic lines on the cone are:*

1. ***Generators****: straight lines radially from the apex.*
2. ***Rolled geodesics****: curves that become straight lines when the cone is developed (unrolled) into a planar sector.*

The developed cone is a circular sector with opening angle $2\pi \sin\alpha \approx 2\pi \times 0.5365 \approx 3.370$ *rad* $\approx 193.1°$.

9 Numerical Examples

Table 1: Cone parameters

Parameter	Exact value	Numerical
Apex (z-coordinate)	$-\frac{3}{4}$	-0.750
Slope m	$\frac{\pi}{2}$	1.5708
Half-angle α (axis–generator)	$\arctan(2/\pi)$	0.5669 rad = 32.48°
Elevation angle β (generator–horiz.)	$\arctan(\pi/2)$	1.0039 rad = 57.52°
Apex angle 2α	$2\arctan(2/\pi)$	64.96°
Base radius ($z = 0$)	$\frac{3}{2\pi}$	0.4775
$\sqrt{1+\pi^2/4}$	$\frac{\sqrt{4+\pi^2}}{2}$	1.8621

Table 2: Lateral surface area: $A = \pi\sqrt{1+\pi^2/4} \cdot (r_2^2 - r_1^2)$

r_1	r_2	A
0	5	146.249
0	10	584.995
5	10	438.746
0	20	2339.979

Table 3: Volume: $V = \frac{4}{3\pi}\left[(h_2 + \frac{3}{4})^3 - (h_1 + \frac{3}{4})^3\right]$

h_1	h_2	V
-0.75	0	0.179
0	10	527.068
-0.75	10	527.247
0	20	3791.601

10 Summary of All Properties

Table 4: Complete summary of cone properties

Property	Value / Formula
Simplified form	$z = -\frac{3}{4} + \frac{\pi}{2}r$
Surface type	Right circular cone
Apex	$(0, 0, -\frac{3}{4})$
Slope	$m = \frac{\pi}{2}$
Half-angle α (axis–generator)	$\arctan(2/\pi) \approx 32.48°$
Elevation angle β	$\arctan(\pi/2) \approx 57.52°$
Apex angle 2α	$2\arctan(2/\pi) \approx 64.96°$
Gaussian curvature	$K = 0$ (developable)
Mean curvature	$H = \dfrac{\pi}{4r\sqrt{1+\pi^2/4}}$
Gradient magnitude	$\lvert\nabla f\rvert = \frac{\pi}{2}$
Lateral surface area	$A = \pi\sqrt{1+\pi^2/4} \cdot (r_2^2 - r_1^2)$
Volume	$V = \frac{4}{3\pi}\left[(h_2 + \frac{3}{4})^3 - (h_1 + \frac{3}{4})^3\right]$

11 Conclusion

This paper has presented a complete mathematical analysis of the cone surface $z = -\frac{3}{4} + \frac{\pi}{2}r$:

1. The complex original expression simplifies elegantly to $z = -\frac{3}{4} + \frac{\pi}{2}r$.
2. The surface is a right circular cone with apex $(0, 0, -\frac{3}{4})$ and slope $\pi/2$.
3. Gaussian curvature $K = 0$ (developable surface, locally isometric to the plane).
4. Cone half-angle (axis–generator): $\alpha = \arctan(2/\pi) \approx 32.48°$; elevation angle: $\beta = \arctan(\pi/2) \approx 57.52°$; apex angle: $2\alpha \approx 64.96°$.
5. Gradient is constant: $|\nabla f| = \pi/2$.
6. Mean curvature: $H = \pi/(4r\sqrt{1+\pi^2/4})$, decreasing as $r \to \infty$.
7. Closed-form expressions exist for lateral surface area and volume.

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